ANGU PRAVEEN MUTHU SWAMY

Computational Engineer

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VISA	Permanent Resident

PROFILE

Computational Engineer with 3+ years of expertise in FEA, thermal, and CFD simulations for complex systems. Proven ability to build automated, scalable simulation workflows using ANSYS, ABAQUS, Python, and MATLAB, achieving up to 96% model complexity reduction. Specialized in multiphysics analysis, structural optimization, and data-driven engineering solutions.

ABAQUS / ANSYS	100%
PYTHON / MATLAB	100%
SU2 / OPENFOAM	90%
CATIA V5 / SPACECLAIM	80%
APDL / VBA / SQL	80%
C / C++	70%

Linux Windows Mac OS GitHub

EDUCATION

10/2017 - 06/2021 **Master of Science, M.Sc** — Technische Universität Braunschweig, Germany
Computational Sciences in Engineering — **Grade: 1.9**

08/2012 - 08/2016 **Bachelor of Engineering, B.E.** — Dayananda Sagar College of Engineering, India
Mechanical Engineering — **Grade: 81%**

PROFESSIONAL EXPERIENCE

09/2025 - Present **Professional Doctorate Candidate** — Amsterdam University of Applied Sciences, Amsterdam

- Developing an Explainable Predictive Maintenance (PdM) prototype using legacy shipboard data to deliver transparent maintenance decisions for marine auxiliary systems.
- Engineering interpretable AI models (GAM, EBM) that integrate explainability metrics directly into operator workflows for verifiable insights.
- Designing a decision-support dashboard to visualize health indices, anomaly indicators, and Remaining Useful Life (RUL) during vessel field trials.

01/2022 - 01/2025 **Computational Engineer** — Capgemini Engineering GmbH, Munich

- Performed 3D thermo-mechanical simulations to evaluate structural integrity and thermal behavior of EUV lithography systems.
- Developed steady-state and transient thermal models using ANSYS, achieving **20% faster** solver convergence.
- Designed and optimized cooling circuits to improve thermal management, yielding a **16% increase** in thermal efficiency.
- Conducted FEM result evaluation, sensitivity studies, and hotspot identification to improve component lifetime and system stability.
- Applied Model Order Reduction (MOR) techniques to reduce matrix size by **96%** while maintaining a numerical accuracy of 10^{-8}.
- Developed MATLAB-based automation tools to optimize encoder testing, reducing processing time by **40%**.
- Authored VBA automation scripts saving **15+ hours/month** and reducing human errors by **90%**.

11/2020 - 06/2021 **Master Thesis** Robert Bosch Automotive Steering GmbH, Schwäbisch Gmünd

- Developed an automated strength verification framework for Electronic Power Steering bolts using ABAQUS and Python, reducing analysis time by **90%**.
- Implemented VDI 2230-compliant joint analysis with $<$**0.5% deviation** from physical tests through Python-based automation.
- Built custom ABAQUS plugins and parametric bolt models enabling standardized and repeatable joint analysis.
- Achieved **3**$\times$ **faster** design iterations by eliminating manual Excel-based workflows.
- Enhanced computational efficiency by **40%** while improving result reliability through experimental validation.

09/2018 - 10/2020 **Scientific Assistant** Open Hybrid Lab Factory GmbH, Wolfsburg

- Performed advanced ANSYS simulations on FRP and hybrid materials, achieving higher accuracy in stress predictions and a **25%** thermal stress reduction through design optimization.
- Engineered crack detection and prediction workflows using ImageJ and Python-based machine learning algorithms, achieving over **80%** prediction accuracy.
- Created VBA-based automation tools reducing data processing time by **50%** across research team workflows.
- Programmed **3 custom UMAT subroutines** for ABAQUS enabling advanced nonlinear material analysis of potting materials.

SCIENTIFIC PROJECTS

2019 - 2020 **Student Thesis** Technische Universität Braunschweig, Germany

- Conducted multi-fidelity aerodynamic optimization of a Blended Wing Body configuration incorporating novel airframe technologies.
- Generated parametric aircraft geometry using OpenVSP with subsequent structured mesh generation via ICEM CFD.
- Performed high-fidelity CFD analysis utilizing the SU2 open-source solver with RANS turbulence modeling.
- Implemented performance estimation for fuel efficiency and maximum takeoff weight using SUAVE's Python-based aircraft sizing algorithms.
- Executed multi-point gradient-based optimization employing MATLAB's Sequential Quadratic Programming (SQP) algorithm.
- Achieved significant aerodynamic improvement through coupled optimization framework validation.
- Increased overall aircraft efficiency by **36%** and reduced drag by **75%**.

LANGUAGES

English - Fluent (IELTS 7.5)
German - Proficient (DSH 2/C1)
Tamil - Mother Tongue
Hindi, Kannada - Fluent

INTERESTS

Swimming
Adventure Sports
Hiking
Motorcycling

CERTIFICATIONS

ANSYS APDL Programming
Deep Learning A-Z™: Hands-On ANN
Complete Modern C++ (C++11/14/17)
Machine Learning by Andrew NG